```
#include <iostream>
using namespace std;
#define size 5
int queue[size];
int front=-1;
int rear=-1;
void enqueue(int val){
if(rear==size-1){
cout<<"overflow"<<endl;
}
else{
if(front==-1){
front=0;
}
rear++;
queue[rear]=val;
cout<<val<<"enqueue"<<endl;
}
}
void dequeue(){
if(front==-1||front>rear){
cout<<"queue is empty"<<endl;
}
else{
cout<<queue[front]<<"removed"<<endl;
front++;
}
}
void peek(){
if(front==-1||front>rear){
cout<<"empty"<<endl;
}
else{

cout<<"peek is"<<queue[front]<<endl;
}
}
```

```
void display(){
if(front==-1||front>rear){
cout<<"queue is empty"<<endl;
}
else{
for(int i=front;i<=rear;i++){
cout<<queue[i]<<endl;
}
}
}
int main(){
enqueue(10);
enqueue(20);
enqueue(30);
display();
dequeue();
peek();
enqueue(40);
dequeue();
dequeue();
display();

return 0;
}
```